Positive Re-wiring of a Dying Brain: Multi-Modal Neuroplasticity Enhancement in Chronic Illness Pain Management

Independent student research paper, reproduced here for reference. Companion literature review to the Neuroplasticity Exploration Simulator. All "reverse/cure" language has been revised to reflect that current evidence supports symptom, biomarker, and progression management — not disease reversal.

Varshith Sekhar

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It is not sudden, like getting splashed in the face by a water balloon on a random evening; it creeps into your own body, without you knowing, and slowly eats you away. The pain caused by chronic illnesses—often due to neurodegenerative disorders such as Parkinson's and Alzheimer's Disease—develops over time and shrinks your brain and life, ultimately making you dysfunctional. In this new age of science, there are ways to combat chronic illnesses and potentially manage or delay their effects. People often think it is a disorder for the elderly; however, there have been prominent cases in individuals as young as 30 developing Parkinson's disease. Most neurodegenerative diseases do not have a definitive cause; in *The Brain's Way of Healing* by Norman Doidge, Parkinson's is described as caused by "the increasing inability of the brain, called the substantia nigra. To produce the brain chemical dopamine, which is necessary for normal movement" (Doidge). For Alzheimer's, a long-known underlying cause is the abnormal build-up of amyloid β-protein plaques and tau tangles ("Alzheimer's Causes and Risk Factors"). This leads to the question: Is it possible to meaningfully manage or delay the effects of chronic illnesses by enhancing neuroplasticity in the brain through a multi-modal approach? Pain management of chronic illness is dominated by short-term pharmaceuticals, but current evidence suggests the effects of chronic illness can be meaningfully reduced or delayed — not reversed or cured — with positive neuroplasticity enhancement using a multi-modal approach of natural and practical methods.

Substantia Nigra and Parkinson's; The progressive degradation of the substantia nigra (SN) is a foundational cause of chronic illness. SN is a dark zone in the brain, in the basal ganglia, playing a vital role in brain chemistry (Cleveland Clinic). It connects to different brain parts to produce dopamine, one of the most important neurotransmitters. The SN and basal ganglia produce around 80% of the body's dopamine (Doidge). Parkinson's disease is one of the most prevalent neurodegenerative/chronic illnesses, with ~60,000 new cases per year (American Association of Neurological Surgeons). It is a movement disorder: small pauses of movement progress to gaits, shuffling, and tremors. Dopamine works with other neurotransmitters to coordinate nerves and muscle cells for movement; no dopamine means uncoordinated movement, resulting in tremor and impaired movement, balance, and coordination (AANS). Dopamine is the main neurotransmitter for movement; a deficit worsens movement—the basis of Parkinson's.

Amyloid Beta Plaques and Tau Tangles; Amyloid-beta (Aβ) plaques and tau tangles—protein build-ups—are chemical indicators of Alzheimer's Disease (AD). Though leading causes of AD, they normally help the brain: Aβ plaques form from Amyloid Precursor Protein that helps with neuron regulation and development (Findley); tau proteins sit within neurons to transport nutrients and molecules (Findley). Excess becomes toxic. Amyloid beta 42 (AB42) develops when abnormal-length amyloid proteins are produced—more length, more clumps, more toxicity (Findley). AB42 is like chunks of protein combining to blockade neural pathways; tau tangles are sticky, thread-like neurofibrillary tangles that prevent tau from transporting nutrients and damage the inner neuron skeleton, impairing cell communication (Findley). There are currently no treatments to decrease these build-ups, but anti-amyloid drugs have been trialed (Findley).

The APOE4 Gene; A less-known risk factor is the APOE4 (apolipoprotein E) gene, carried by ~25% of people; ~2–3% have two copies. The stronger the presence of APOE4, the worse AD becomes (Bryant). The mechanism is still poorly understood but resembles

AB42/tau behavior: the gene causes a build-up of cholesterol in the brain (Bryant). APOE normally carries cholesterol and important fats to the bloodstream, but APOE4 produces an excess the brain can't process fast enough, causing lipid/cholesterol imbalances that decrease cell-membrane creation, molecule transport, and energy—all vital to combatting AD. Few treatments exist; a sister gene APOE2 (rare) provides little protection (Bryant). Choline supplementation has been used to mitigate the metabolic imbalances caused by APOE4 and increase neuroplasticity of genetically-caused chronic illness.

Neuroplasticity; Neuroplasticity = the brain (neuro) rewiring itself (plasticity): "Neuroplasticity refers to the brain's ability to change and adapt throughout life by modifying its structure, functions, or neural pathways" (Fisher). It doesn't necessarily mean the brain can heal itself. It can be enhanced negatively or positively—strengthening neuron connections for a specific action. For chronic illness, pain-sense neurons strengthen every time they fire together, because *neurons that fire together, wire together* (Doidge 16). But with MIRROR—Motivation, Intent, Relentlessness, (self) Reliability, Opportunity, and Restoration—a patient can turn neuroplasticity into positive rewiring and cause healing (Doidge). Positive neuroplasticity can rewire the same pain-sense neurons to reduce pain and may help manage chronic-illness effects. "The brain has the amazing ability to rewire itself so that we can learn new skills, function better, decrease chronic pain, and even regain functioning after a stroke or traumatic brain injury" ("Neuroplasticity Training: A Missing Piece to Chronic Illness").

Multi-Modal Approach; Aerobic Exercise; Aerobic exercise increases blood flow to the brain. In chronic (especially neurological) illness, the brain degrades and becomes "suffocated," with no space for neurogenesis or synaptogenesis. Aerobic exercise opens brain blood vessels, "expands the brain," and increases growth factors (Fisher). In AD the hippocampus shrinks, hurting memory retention; aerobic exercise increases hippocampal volume and "increases blood flow to the brain and reduces stress and inflammation. Together these changes improve mood, memory, focus and processing speed" (Fisher). In Doidge's novel, patient John Pepper has had Parkinson's since his thirties yet has no visible symptoms at 77 because he exercises: "Because PD is a movement disorder…the more I move, the slower PD would be able to take over my life" (Doidge). A rat experiment: "Those raised under normal lab conditions, without much exercise developed Huntington's disease—a worsened form of PD—as expected. Those that got a lot of fast walking and stimulation got the disease too, but onset was significantly delayed" (Doidge). "Physical exercise…increases brain derived neurotrophic factor (BDNF)—important neuron survival and growth" ("Neuroplasticity Training").

Diet; A good diet rich in certain proteins and omega fatty acids increases brain function, cognitive ability, and biomarkers. W-3 PUFAs are "therapeutic agents because of their anti-inflammatory, antioxidant, and neurogenic promoting activities" (Chávez-Castillo). W-3-rich diets show "lower risk of the development and progression of cognitive impairment" because "PUFAs promote the disruption of the P38 of mitogen activated protein kinase…involved in the biosynthesis of pro-inflammatory cytokines like TNF-α and IL-1β" (Chávez-Castillo). DHA and EPA support synaptic plasticity: "Both DHA and EPA are part of the lipid membrane and act as substrates for molecule synthesis, improving cognitive processes and performing neuroprotective activities" (Chávez-Castillo). Intermittent fasting "promotes improved cellular functioning by way of reducing inflammation and oxidative stress and increasing cellular metabolism…increases synaptic plasticity" (Sibille).

Mental Health: CST and Meditation; Keeping a calm mind matters as much as exercise and diet. Meditation and CST increase mood, cognitive function, and dopamine levels. The basal ganglia accounts for ~80% of dopamine; a deficit (SN damage) can start Parkinson's. Consistent negative emotions decrease dopamine; meditation calms the brain, potentially increasing dopamine and helping manage Parkinson's symptoms. In fibromyalgia patients, "body awareness interventions including mindfulness meditation, guided imagery, relaxation programs, yoga, and hypnotherapy had positive effects on fibromyalgia impact, pain, anxiety, depression, and quality of life" ("Neuroplasticity Training"). "Meditation is believed to support neuroplasticity by fostering the growth of new brain cells and connections" (Fisher). CST is "exercise therapy" for the brain; in AD it works against weakened memory retention through puzzles and crosswords: "CST may thus improve cognitive function associated with CR by strengthening neural functional connectivity in the AD patients' brain" (Ubuka).

Visualization of Brain Maps; A new method: visualizing brain maps—imagining the neurons and pain maps responsible for pain to disappear, weaken, or rebuild. In Doidge's novel, a neurologist with years of pain from a spinal-cord injury imagined his pain maps: "Each time he got an attack of pain, he immediately began visualizing…the very brain maps he had drawn…'disconnect the network, shrink the map'" (Doidge), eventually lessening the pain for years. In a back-pain study, scientists "observed that visual feedback of one's own back reduces the perceived intensity of acute painful stimuli applied to it," suggesting "the alteration of the body image or visual feedback of the painful region…may decrease pain in chronic pain patients" (Diers). "When you visualize something, the same neural pathways are engaged in the brain as when you actually perform the action"; "Visualization allows for neuroplasticity to take place; it strengthens connections of motor and cognitive pathways" ("The Science Behind Visualization"). "Consistency in visualization practice can be just as important as physical training."

Limitations; People struggle to find time for non-pharmacological chronic-pain management; many have trauma responses. A 9–5 (or longer) leaves little time for exercise, meditation, and diet planning—methods requiring consistency, dedication, relentlessness (MIRROR). For most patients, a pill is more efficient. Chronic illness can trigger a trauma/fear response inhibiting these practices (e.g., a spinal-cord-injury patient fears worsening the injury during spinal exercises), strengthening pain-sense neurons. But research shows pharmaceuticals only manage symptoms temporarily (Doidge 12), and non-invasive natural methods can be paired with pharmaceuticals to enhance treatment further ("Neuroplasticity Training").

Conclusion; Managing and delaying chronic illness and its pain is not found only through pharmacological treatment; many methods alter the brain's functioning, healing, and rewiring (neuroplasticity). Cognitive decline has been treated by "non-pharmacological approaches that increase CR and mitigate cognitive decline by stimulating neuroplasticity in the brain" (Ubuka). Even ancient methods—exercise and proper diets, accessible to everyone—increase biomarkers like BDNF, GDNF, and neurogenesis. Being happy can increase dopamine; visualizing your brain healing can mitigate chronic pain. Chronic illness is not a surprise water-balloon attack—it is slowly intrusive. But managing and delaying it is possible by practicing scientific methods already available; it is having the self-belief and grit to want to heal and continue to heal.

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